

# Angel Koradiya

koradiyaangel11@gmail.com | +91 88668 39115 | Ahmedabad, Gujarat, India  
linkedin.com/in/angel-koradiya | github.com/ANGELKORADIYA | angelkoradiya.in

## Professional Summary

AI/ML & Full-Stack Engineer with ~2 years of experience architecting production GenAI platforms, natural-language-to-SQL engines, and high-concurrency cloud services across Python and TypeScript. Proven track record of engineering scalable RAG architectures, local-first search systems, and automated enterprise pipelines with robust AST-based algorithmic security.

## Professional Experience

### SilverTouch Technologies Ltd.

AI/ML Engineer (GenAI & Full-Stack)

Ahmedabad, India

Jan 2025 – Aug 2026

- **Talk2DB (Natural-Language-to-SQL Platform):** Engineered enterprise text-to-SQL engine on Vanna.AI, Qdrant, and self-correcting Ollama retry loops, achieving **85% execution accuracy** across 40+ relational tables; designed AST-based scope injection with SQLGlot to enforce role-based access control and block prompt jailbreaks.
- **AeroBatch (Intelligent Document AI):** Architected an on-premises document AI pipeline (PaddleOCR, Celery, FastAPI, vLLM) processing 100+ page specifications at **2 pages/min on CPU**; constructed knowledge graphs across 20+ fields with confidence fusion to eliminate hallucinations.
- **Financial Reconciliation Engine (Bank-Reco):** Architected pipeline with 10 matching strategies (RapidFuzz, combinatorial subset-sum), reconciling 1,200 SAP entries against 4,500 bank rows at **80–99% precision**, cutting reconciliation cycle from **15 days to under 30 minutes**.
- **AI Copilot & Autonomous Workflow Builder:** Led migration to a self-hosted multi-tenant Qdrant cluster; built an interactive drag-and-drop DAG canvas (Next.js, TypeScript, WebSockets) translating natural language requests into executable backend operations.

## Key Projects

### OmniQuery | TypeScript, FastAPI, React, Qdrant, PostgreSQL

[Source Code]

- Built an enterprise NL-to-SQL platform that auto-generates training pairs enriched with distinct DDL column values, reducing ad-hoc database query turnaround time by **80%**.
- Enforced dual-layer RBAC validating native PostgreSQL permissions, generating role-partitioned Qdrant collections to guarantee complete multi-tenant schema isolation.

### IndexForge | Rust, Axum, Tantivy, pgvector, Tree-sitter

[Source Code]

- Developed a compiled 9-crate hybrid search engine in Rust fanning out 3 parallel search arms (BM25, trigram fuzzy, HNSW vector) with **sub-10ms p50 latency** across 500K+ chunks.
- Merged candidate distributions via Reciprocal Rank Fusion (**RRF**,  $k = 60$ ) and cross-encoders; integrated Tree-sitter for language-aware AST code chunking.

### ForgeOAgent | Python, FastAPI, Gemini API, wxPython, LangGraph

[Live Demo] | [Source Code]

- Built a four-interface platform (PyPI, CLI, desktop GUI, web API) with an autonomous orchestrator that dynamically decomposes complex goals and coordinates specialized sub-agents.
- Engineered dynamic AST-based tool reflection from raw Python classes and runtime pip auto-resolution, executing sandboxed tasks with **sub-2s reasoning cycles**.

## Technical Skills

**Programming Languages:** Python (async), TypeScript, JavaScript, SQL (PostgreSQL, MySQL, SQLite), Rust, C/C++, Java.  
**AI, ML & GenAI:** RAG, Agentic Workflows (LangGraph), Vector DBs (Qdrant, pgvector), Ollama, Vanna.AI, SQLGlot, OCR, Whisper, vLLM, Scikit-Learn, PyTorch.  
**Backend & Full-Stack:** FastAPI, Next.js, Node.js, React, PostgreSQL, MongoDB, Redis, WebSockets/SSE, Celery, GraphQL.  
**DevOps & Core CS:** Docker, Git, Linux, CI/CD, System Design, Data Structures & Algorithms (DSA), OOP, OS, DBMS.

## Education & Academic Credentials

### Vishwakarma Government Engineering College (VGEC)

Bachelor of Engineering in Computer Engineering (CGPA: **8.01 / 10.0**)

Ahmedabad, India

2021 – May 2025

National Examinations: **GATE 2025 Qualified on 1st Attempt** | **JEE MAIN: 93.5 Percentile**